Mathematical Specification: Non-Additive Shadow-Gravity N-Body Model

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This is the frozen mathematical contract for a classical N-body simulation of non-relativistic shadow gravity with the full non-additive attenuated-ray force law. Any ambiguity found during implementation is resolved by the governing definitions (eqs. 3, 9, 16, 17), not by implementer discretion.

Conventions

- Code units with G_N as an input (default G_N = 1). Arithmetic is IEEE binary64; use expm1 and log1p wherever specified.
- Integrals over the sphere use dΩ. The unit vector n̂ points toward the flux source, so a corpuscle travels along −n̂.
- For target k and occluder j: c_kj = x_j − x_k, d_kj = |c_kj|, ĉ_kj = c_kj / d_kj.
- Equation numbers are those of the specification; numbers 14, 15 and 18 are intentionally unused.

1. State vector and system definition

Every particle is a finite-size uniform sphere for optical-depth purposes, and the domain is open with an optional replenishing background extinction.

1.1 Stored per particle (i = 1..N)

- Position x_i and velocity v_i in R^3.
- Inertial mass m_i > 0 and bulk density ρ_i > 0.
- Species offset δ_i (default 0).
- Type flag: body or probe.

1.2 Derived per particle

Recompute whenever m_i, ρ_i, δ_i or the global κ_0 change.

$$a_i=\Big(\tfrac{3m_i}{4\pi\rho_i}\Big)^{1/3},\quad \kappa_i=\kappa_0(1+\delta_i),\quad \beta_i=\kappa_i\rho_i,\quad \tau_{c,i}=2\beta_ia_i=\frac{3\kappa_im_i}{2\pi a_i^2}\tag{1}$$

$$\mathcal{A}(\tau)=1-\frac{2}{\tau^{2}}\big[1-(1+\tau)e^{-\tau}\big],\qquad A_i^{\rm eff}=\pi a_i^2\,\mathcal{A}(\tau_{c,i}),\quad S_i=\frac{3\,\mathcal{A}(\tau_{c,i})}{2\tau_{c,i}},\quad m_i^{g}=(1+\delta_i)\,m_iS_i\tag{2}$$

For $\tau < 0.05$ evaluate the series below to avoid cancellation.

$$\mathcal{A}=\tfrac{2\tau}{3}-\tfrac{\tau^2}{4}+\tfrac{\tau^3}{15}-\tfrac{\tau^4}{72}+\tfrac{\tau^5}{420}-\tfrac{\tau^6}{2880}$$

This gives $S = 1 - (3/8)\tau + (1/10)\tau^2 - \dots$ and $A_{eff} = \kappa_0 m^g$. In the thick limit $A_{eff} \to \pi a^2$.

1.3 Global state

- Time t and step h.
- κ_0 (mass extinction coefficient), momentum-transfer ratio ξ, corpuscle speed v_g.
- Energy-deposition factors ζ and η_E.
- Background extinction $\alpha_{bg} \geq 0$ and flux rest-frame velocity W (default 0).
- Velocity mode M in {V0, V1, V2}.
- Quadrature parameters ($N_{side},0$, ℓ_{cap}, f, n_{GL}).
- Derived momentum flux Π (eq. 19).

1.4 Geometric model

- **Occluders.** Each body is a uniform-density sphere of radius a_i and extinction coefficient β_i (1/length). Overlapping spheres have additive β.
- **Targets.** Each body is point-sampled at its center for incoming-flux attenuation. Its momentum-absorbing area is $A_{eff,k}$, which includes self-shielding.
- **Probes.** Point targets with unit mass and $A_p = \kappa_0$, no self-shielding. They never occlude and are not integrated.
- **Scattering.** Scattered flux is removed from the beam and not re-tracked (single-removal model). The coupling ξ carries the momentum bookkeeping.

1.5 Domain

The domain is open and unbounded; periodic domains are not supported. Ambient intensity is $I_0 = \Pi/4\pi$ per steradian, isotropic and stationary. The ambient flux is replenished by the background medium. Along any ray, with s measured toward the target:

```
\frac{dI}{ds}=-\alpha_{\rm bg}(I-I_0)-\rho_{\rm ext}I
```

With α_bg = 0, $I = I_0\, e^{-\tau}$. With α_bg > 0, deficits relax to I_0 over 1/α_bg (Seeliger screening). Empty space returns exactly I_0, so there is no confining force at any α_bg.

1.6 Edge cases

- N = 1: force is 0.
- d_kj < 1e-12 a_j: pair term = 0, direction undefined, disc θ_j = π, D_kj evaluated with L = a_j.
- m ⩽ 0 or ρ ⩽ 0 is rejected.
- NaN or Inf in any force aborts the step.

2. Force law

The force is the exact attenuated-ray integral, coded in deficit form. Never integrate the transmission 🚫 directly: that causes catastrophic cancellation at small τ.

2.1 Master expression (mode V0)

Let D_k(n̂) in [0, 1) be the flux deficit at x_k from direction n̂ (section 4). Using the integral of n̂ over the sphere = 0:

```
\mathbf F_k^{\rm sh}=-\frac{\xi\Pi A_k^{\rm eff}}{4\pi}\int \hat{\mathbf
n}\,\mathcal T_k\,d\Omega=\Gamma_k\int_{\mathbb S^2}\hat{\mathbf
n}\,D_k(\hat{\mathbf n})\,d\Omega,\qquad \Gamma_k=\frac{\xi\Pi A_k^{\rm eff}}
{4\pi},\quad \mathcal T_k=1-D_k\tag{3}
```

The acceleration is a_k = F_k / m_k.

2.2 Exact decomposition (coded form)

Let D_kj be the deficit if only occluder j existed. Rays that intersect at most one sphere satisfy D_k = Σ_j D_kj exactly, so:

$$\mathbf F_k^{\rm sh}=\Gamma_k\Big[\sum_{j\ne k}P_{kj}\,\hat{\mathbf c}_{kj}+\sum_{c\in\mathcal L_k}A_{\ell(c)}\,\hat{\mathbf n}_c\Big(D_k(\hat{\mathbf n}_c)-\sum_{j\in T_c}D_{kj}(\hat{\mathbf n}_c)\Big)\Big]\tag{4}$$

The first sum is the pair-exact single-occluder force, including saturation and screening. The second is the overlap correction, nonzero only where rays intersect two or more spheres. L_k, T_c and A_ℓ are defined in section 4.3.

2.3 Pair function

Let θ_j = arcsin(a_j / d_kj) for d_kj > a_j, and θ_j = π otherwise. Axisymmetry about ĉ_kj gives P_kj = 2π ∫ D_kj cosθ′ sinθ′ dθ′. Define λ = α_bg + β_j and the single-sphere deficit:

$$D_{kj}=e^{-\alpha_{\rm bg}\max(0,s_1)}\,\frac{\beta_j}{\lambda}\big(-\mathrm{expm1}(-\lambda L)\big)\quad(\lambda>0),\qquad D_{kj}=0\ (\lambda=0)\tag{5}$$

Here L and s_1 are the chord length and near-entry distance (eq. 16).

Exterior (d > a_j). With b = a_j sinφ, L = 2 a_j cosφ, s_1 = sqrt(d² − a_j² sin²φ) − a_j cosφ:

$$P_{kj}=2\pi\frac{a_j^2}{d_{kj}^2}\int_0^{\pi/2}D_{kj}(\varphi)\sin\varphi\cos\varphi\,d\varphi\tag{6a}$$

For α_bg = 0 this is exactly P_kj = A_eff,j / d_kj² for every τ_c (not an approximation), because cosθ′ dΩ = dA_perp / d². Use this closed form. Otherwise use n_GL-point Gauss-Legendre in φ.

Interior or contact (d ⩽ a_j). With μ = cosθ′, L = dμ + sqrt(a_j² − d²(1 − μ²)) and s_1 < 0:

$$P_{kj}=2\pi\int_{-1}^{1}D_{kj}(\mu)\,\mu\,d\mu\tag{6b}$$

Evaluate by composite Gauss-Legendre on the panels μ in {−1, −0.5, −0.1, −0.02, 0, 0.02, 0.1, 0.5, 1} with 16 points per panel.

2.4 Thin-limit reduction

As τ_c → 0: A_eff,k → κ_k m_k, D_kj → β_j L, P_kj → κ_j m_j / d² (exterior) or κ_j m_j d / a_j³ (interior), the overlap term is O(τ²), and S → 1. Therefore:

$$\mathbf F_{kj}\to\frac{\xi\Pi\kappa_k\kappa_j}{4\pi}\frac{m_km_j}{d^2}\hat{\mathbf c}=G_N(1+\delta_k)(1+\delta_j)\frac{m_km_j}{d^2}\hat{\mathbf c}\quad\text{when}\quad \Pi=\frac{4\pi G_N}{\xi\kappa_0^2}\tag{7}$$

The interior thin result is exactly the uniform-sphere field, so finite size supplies the softening and no separate Plummer length is added.

2.5 Velocity-dependent terms

Define $\varepsilon_k = (v_k - W) / v_g$, $e_k = |\varepsilon_k|$, and $g(\hat n;\varepsilon) = |\hat n + \varepsilon|\,(\hat n + \varepsilon)$.

- **V0:** static, $F = F_{sh}$.
- **V1 (ambient drag):** $F = F_{sh} + F_{dr}$.
- **V2 (aberration, full shielding):** $F = F_{sh} + F_{dr} + F_{ab}$. Requires $\alpha_{bg} = 0$ and $\max|\varepsilon| \leqslant 0.1$.

$$\mathbf F_k^{\rm dr}=-\frac{\xi\Pi A^{\rm eff}_k}{4\pi}\int\mathbf g\,d\Omega=-\xi\Pi A_k^{\rm eff}H(e_k)\hat{\boldsymbol\epsilon}_k,\quad H(e)=\begin{cases}\tfrac43e+\tfrac4{15}e^3&e\le1\\[2pt] e^2+\tfrac23-\tfrac1{15e^2}&e>1\end{cases}\tag{8}$$

Eq. 8 is exact in e and unshielded, so it is valid only when the sky-covering fraction is much less than 1. The flux frame is the simulation frame shifted by W (Galilean).

$$\mathbf F^{\rm ab}_k=\Gamma_k\int\Big[\mathbf g(\hat{\mathbf n};\boldsymbol\epsilon_k)\,D^{(\epsilon)}_k(\hat{\mathbf n})-\hat{\mathbf n}\,D_k(\hat{\mathbf n})\Big]d\Omega\tag{9}$$

Here $D_k^{(\varepsilon)} = -\mathrm{expm1}(-\Sigma_j \beta_j L_j(\hat q_j))$ with $\hat q_j = (\hat n + \varepsilon_j) / |\hat n + \varepsilon_j|$, and $L_j(\hat q)$ is the chord of sphere j along the ray from x_k in direction $\hat q$ (eq. 16). This is the retarded-occluder geometry: the hole a moving sphere leaves in the stream is a tube with axis $\hat q_j$.

Eq. 9 is evaluated by the adaptive scheme of section 4.3, with the touching discs replaced by the union of static discs and shifted discs (center = image of $\hat c_{kj}$ under $\hat q \to \hat n$, radius $\theta_j / (1 - |\varepsilon_j|)$). The leaf contribution is $A_\ell\,[g_c\,D^{(\varepsilon)} - \hat n_c\,D]$.

First-order reference for thin point-like pairs. This must be verified against the quadrature, which is the governing definition:

$$\mathbf F^{\rm sh}_{kj}+\mathbf F^{\rm ab}_{kj}\simeq G_N\frac{m_k^gm_j^g}{d^2}\Big\{\hat{\mathbf c}\big[1+\hat{\mathbf c}\cdot(3\boldsymbol\epsilon_j+\boldsymbol\epsilon_k)\big]+(\boldsymbol\epsilon_k-\boldsymbol\epsilon_j)\Big\}\tag{10}$$

2.6 Singularity handling

There is no $1/r^2$ singularity. Finite-size spheres give $P_{kj} \to 0$ as $d \to 0$ (interior form), and a_i is derived from ρ_i, so a point mass is excluded by construction (shrinking a at fixed κm drives τ_c to infinity). Overlapping bodies are permitted and the force remains finite and continuous at $d = a_j$.

3. Numerical integration

Velocity-Verlet (kick-drift-kick) is the required scheme. It is second-order and needs one force evaluation per step in V0. It is symplectic for the pairwise-conservative thin regime, keeps angular momentum exact for central pair forces, and keeps energy errors bounded at $O(h^2)$. Outside that regime (non-additive, dissipative) the force is non-conservative and no integrator preserves energy, so energy drift there is physical.

3.1 Update equations

$$\mathbf v_{n+\frac12}=\mathbf v_n+\tfrac h2\mathbf a_n,\qquad \mathbf x_{n+1}=\mathbf x_n+h\mathbf v_{n+\frac12},\qquad \mathbf a_{n+1}=\mathbf a(\mathbf x_{n+1}),\qquad \mathbf v_{n+1}=\mathbf v_{n+\frac12}+\tfrac h2\mathbf a_{n+1}\tag{11}$$

3.2 Velocity-dependent modes (V1, V2)

Replace the last two updates with the fixed-point solve:

$$\mathbf v^{(0)}=\mathbf v_{n+\frac12}+\tfrac h2\mathbf a_n,\qquad \mathbf v^{(i+1)}=\mathbf v_{n+\frac12}+\tfrac h2\,\mathbf a(\mathbf x_{n+1},\mathbf v^{(i)})\tag{12}$$

Iterate until $\max_k |v^{(i+1)} - v^{(i)}| / (|v^{(i+1)}| + 1e\text{-}30) < 1e\text{-}12$ or 6 iterations. Set $v_{n+1} = v^{(i+1)}$ and store the last evaluated a_{n+1}. Contraction requires $h\gamma$ much less than 1 (see 3.3).

3.3 Time step

The default is a fixed h (tests, reversibility). The adaptive (interactive) mode uses a global step recomputed each step:

```
h=\min\Big\{h_{\max},\ \eta_t\min_k\sqrt{\ell_k/|\mathbf a_k|},\
\eta_t\min_k\ell_k/|\mathbf v_k|,\ \frac{0.1}{\max_k\gamma_k}\Big\},\quad
\ell_k=\max\big(a_k,\min_jd_{kj}\big),\ \eta_t=0.005\tag{13}
```

Here γ_k = |F_dr,k| / (m_k |v_k − W|). Adaptive steps break time-reversibility and symplecticity, and results obtained with them must be flagged as such.

3.4 Conservation monitoring

Report every step:

- Energy E = Σ ½ m_k v_k² + Σ_{k<j} U_kj, with U_kj = −G_N m^g_k m^g_j / d for d ⩾ a_j, and inside the sphere U_kj(d) = U_kj(a_j) − ∫_d^{a_j} Γ_k P_kj(r) dr.
- Linear momentum P = Σ m_k v_k and angular momentum L = Σ m_k x_k × v_k.
- Drift measures |ΔE| / |E$_0$|, |ΔP| / Σ m|v|, |ΔL| / |L$_0$|, each with its running maximum.
- Momentum-balance residual R_P = |Σ_k F_k| / Σ_k |F_k|.

E is a valid conserved quantity only when pair forces are reciprocal (V0, α_bg = 0, non-overlapping or equal-radius). Otherwise label it "diagnostic, non-conservative regime".

For V0 with α_bg = 0 and an isolated flux-absorbing set, Σ_k F_k = 0 exactly in the model (opposite rays have equal τ), so R_P measures quadrature and center-sampling error. For α_bg > 0 it is physically nonzero, because the background absorbs momentum.

4. Optical-depth and ray evaluation

4.1 Ray-sphere intersection

For the ray x_k + s q̂ (s > 0) and sphere j with c = x_j − x_k:

```
t=\mathbf c\cdot\hat{\mathbf q},\quad b^2=|\mathbf c|^2-t^2,\quad
h_j=\sqrt{a_j^2-b^2}\ (b<a_j),\quad s_{1,2}=t\mp h_j,\quad L_j=\max(0,s_2)-
\max(0,s_1)\tag{16}
```

Spheres with $b \geq a_j$ or $s_2 \leq 0$ are skipped. If x_k is inside sphere j, $s_1 < 0 < s_2$ and $L_j = s_2$. Self-occlusion (j = k) is excluded. Optical depth is $\tau_j = \beta_j L_j$, and $\tau(\hat{n}) = \Sigma_j \tau_j$ is exactly additive.

4.2 Deficit along a ray

- **$\alpha_{bg} = 0$ (all modes):** $D = -\mathrm{expm1}(-\Sigma_j \tau_j)$.
- **$\alpha_{bg} > 0$ (V0 and V1 only):** sweep from far to near. Gather the intervals $[s_{1,j}, s_{2,j}]$ with $s_{2,j} > 0$. The breakpoints are all interval endpoints, sorted descending. Start at $D = 0$ at the largest s_2. For each elementary segment of length ℓ, let β_{seg} be the sum of β_j of the covering spheres (recompute per segment, or use compensated summation) and $\lambda = \alpha_{bg} + \beta_{seg}$:

$$D\leftarrow\frac{\beta_{\rm seg}}{\lambda}\big(-\mathrm{expm1}(-\lambda\ell)\big)+D\,e^{-\lambda\ell}\quad(\lambda>0)\tag{17}$$

After the last segment apply the gap factor $D \leftarrow D\, e^{-\alpha_{bg} s_{near}}$, where s_{near} is the smallest positive interval endpoint. This reduces to the $\alpha_{bg} = 0$ form as $\alpha_{bg} \to 0$, and gaps are segments with $\beta_{seg} = 0$.

4.3 Quadrature of the overlap correction

1. **Discs on the sky of k.** Disc j has center $\hat{c}_{kj}$ and angular radius θ_j (section 2.3).
2. **Overlap graph.** Nodes are occluders $j \neq k$. Edge (j, l) exists iff $\arccos(\hat{c}_{kj} \cdot \hat{c}_{kl}) < \theta_j + \theta_l$. Only nodes of degree ≥ 1 form the candidate set C_k. Isolated discs cannot contribute to the correction (exact).
3. **Cell structure.** HEALPix nested scheme, base $N_{side,0} = 8$ (768 cells). Level ℓ has area $A_\ell = \pi / (3 N_{side}^2)$ with $N_{side} = N_{side,0} \cdot 2^\ell$, and a conservative bounding radius $\rho_\ell = 1.5 \sqrt{A_\ell}$.
4. **Touching set.** $T_c = \{ j \in C_k : \psi_{cj} - \theta_j \leq \rho_\ell \}$, where $\psi_{cj} = \arccos(\hat{n}_c \cdot \hat{c}_{kj})$, inherited from the parent cell.
5. **Cell rule.** If $|T_c| \leq 1$, the cell contributes zero (exact). Otherwise let $\theta_{eff,j} = \min(\theta_j, 1)$ and θ_{min} = min over T_c of $\theta_{eff,j}$. If $\rho_\ell \leq f\, \theta_{min}$ or $\ell = \ell_{cap}$, the cell is a leaf in L_k. Otherwise it splits into four children.
6. **Leaf term.** $A_\ell\, \hat{n}_c\, [D_k(\hat{n}_c) - \Sigma_{j \in T_c} D_{kj}(\hat{n}_c)]$ (midpoint rule, vector weight $A_\ell\, \hat{n}_c$).

Defaults are f = 0.15 and ℓ_cap = 12 (ρ about 1e-5 rad). Edge singularities of the square-root profile give O(h^(3/2)) midpoint error on cells containing a silhouette; the accuracy target is relative force error of at most 1e-4 in the correction. Any leaf at ℓ_cap with ρ_ℓ > f θ_min sets a cap-hit flag.

4.4 Data structure and complexity

- **Pair part.** Θ(N²), closed form per pair in the default regime. Gauss-Legendre pairs (α_bg > 0 or interior) cost O(n_GL) each.
- **Overlap graph per target.** A spatial hash of discs on a coarse HEALPix grid: O(N log N) typical, O(N²) worst case, giving O(N² log N) per step.
- **Correction per target.** O(|L_k| · m̄), with |L_k| about O(f⁻²) per overlapping pair and m̄ the mean touching-set size.
- **Reference mode.** A brute-force all-rays mode, O(N³), is required for validation at N ⩽ 50.
- **Design target (not guaranteed).** N up to about 500 interactive in V0 on a multicore CPU; N up to about 5000 for mostly disjoint discs on a GPU; dense overlap up to about 300.
- **Thin-fast mode.** Barnes-Hut or FMM is permitted only in a labeled "thin-fast" mode (V0, α_bg = 0, all τ_c < 1e-2, uniform-sphere pair field, opening angle at most 0.5). That mode is non-conforming.

5. Free parameters and calibration

The calibration relation below sets Π so that G_eff = G_N in the thin limit.

```
\boxed{\ \Pi=\frac{4\pi G_N}{\xi\,\kappa_0^{2}}\ }\tag{19}
```

Only κ_0, ξ and v_g are independently settable. Π is derived and read-only, except in "uncalibrated mode", where the code displays G_eff / G_N = ξ Π κ_0^2 / (4π G_N). Changing κ_0 live automatically rescales Π, so the thin-limit orbit is unchanged while τ_c, drag and heating change. At fixed calibration ξ drops out of forces and drag (ξΠ is invariant), but heating depends on Π alone.

Symbol	Meaning	Unit	Default	Range	Live
G_N	calibration target	code	1	fixed	no

Symbol	Meaning	Unit	Default	Range	Live
κ_0	mass extinction coefficient	L^2/M	2e-7	1e-12 to 1e-2	yes
ξ	momentum-transfer ratio	none	1	0.5 to 2	yes
v_g	corpuscle speed	L/T	1e3	10 to 1e8	yes
ζ	energy per momentum factor	none	0.5	0.5 or 1	yes
η_E	deposited fraction	none	1	0 to 1	yes
α_bg	background extinction	1/L	0	0 to 1/L_sys	yes
W	flux frame velocity	L/T	0	magnitude of W at most 0.1 v_g	yes
M	velocity mode	none	V0	V0, V1, V2	yes
m_i	inertial mass	M	1	greater than 0	yes
ρ_i	bulk density	M/L^3	2.39e5 (a = 0.01 at m = 1)	a in 1e-4 to 0.5	yes
δ_i	species offset	none	0	0 to 1e-3	yes
N_side,0	base HEALPix resolution	none	8	4 to 16	no
ℓ_cap	refinement cap	none	12	6 to 14	no
f	refinement tolerance	none	0.15	0.05 to 0.5	yes
n_GL	Gauss-Legendre points	none	32	16 to 64	no
h or η_t, h_max, tol	time step controls	various	fixed h; η_t = 0.005	n/a	yes

The default κ_0 gives τ_c of about 1e-3. Π at defaults is about 3e14 code pressure units. Anisotropic flux is excluded from this frozen model.

6. Diagnostic quantities

All diagnostics are computed with the same quadrature as the force.

1. **Effective G.**
   - Pair: $G_{eff,kj}(d) = |\Gamma_k P_{kj}| d^2 / (m_k m_j)$. For $d > a_j$ and $\alpha_{bg} = 0$ the expected value is $G_N (1+\delta_k)(1+\delta_j) S_k S_j$.

- Many-body: fixed-position probes give G_eff(r) = |g_p| r² / Σ m^g_i, with g_p = Γ_p ∫ n̂ D_p dΩ.

2. **Newtonian residual.**
   - R^N_k = |F_k − F^N_k| / |F^N_k|, where F^N_k = G_N m_k (1+δ_k) Σ_j (1+δ_j) m_j ĉ / d² (unsaturated point masses).
   - R^thin_k is the same residual against the uniform-sphere thin field.
3. **Non-additivity.**
   - Δ^NA_k = |F^sh_k − Γ_k Σ_j P_kj ĉ_kj| / |F^sh_k| (multi-occluder shielding only).
   - Δ^sat_k = |Σ_j Γ_k P_kj ĉ_kj − F^thin_k| / |F^thin_k| (single-body saturation).
4. **Heating** (V0 geometry, static target):

$$\mathcal P^{\rm heat}_k=\eta_E\zeta\,\Pi\,v_g\,A_k^{\rm eff}(1-\bar D_k),\qquad \bar D_k=\frac1{4\pi}\Big[\sum_j\Omega_{kj}+\sum_{c}A_\ell\big(D_k-\textstyle\sum_{T_c}D_{kj}\big)\Big]\tag{20}$$

In eq. 20, Ω_kj = 2π ∫ D sinθ′ dθ′. Exterior form: 2π (a²/d²) ∫₀^(π/2) D sinφ cosφ / sqrt(1 − (a/d)² sin²φ) dφ. Interior form: 2π ∫ from −1 to 1 of D dμ. The isolated-thin reference is P_heat / m = 4π G_N η_E ζ v_g / (ξ κ₀). Also report total power.

5. **Drag rate.** γ_k = |F_dr,k| / (m_k |v_k − W|). Reference for e much less than 1: γ = 16π G_N (1+δ_k) S_k / (3 κ₀ v_g).
6. **Optical depth.**
   - τ̄_k = (1/4π) ∫ τ_k dΩ = Σ_j (1/4π) ∫ τ_kj dΩ (additive, same pair quadrature).
   - τ_max,k = max over {ĉ_kj} ∪ {n̂_c}. This is a lower bound.
   - Per-particle τ_c,i and S_i, plus global min, mean and max.
7. **Sky coverage.** D̄_k per particle, mean and max.
8. **Seeliger check.** Ratio F_kj(α_bg) / F_kj(0) against exp(−α_bg d) for small a_j.
9. **Equivalence-principle measure.** η_EP = 2 |a_k − a_j| / (|a_k| + |a_j|) for test bodies differing in δ.
10. **Conservation (section 3.4).** E, P, L drifts and R_P.
11. **Quadrature health.** Leaf count, cap-hit count, max |correction| / |F|, and brute-force comparison error in reference mode.
12. **Validity flags.** Thin-valid (max τ_c < 0.1), V2-valid (max |ε| ⩽ 0.1), overlap present, interior target present.

7. Initial conditions and test problems

All tests use code units with G_N = 1. Tests (a) to (e) must pass at the stated tolerances before the code is considered conforming.

(a) Two-body circular orbit (thin limit)

Setup. $m_1 = m_2 = 1$, a = 0.01 ($\rho = \rho_0$), $r_0 = 1$, τ_c = 1e-6 (κ_0 = 2.094e-10), V0, $\alpha_{bg} = 0$, fixed h = T/1000, 100 orbits. Set the relative speed v = sqrt(r_0 F / μ) using the model's own radial force.

Expected. S = 1 − 3τ_c/8 = 1 − 3.75e-7, so G_eff / G_N = S^2, ω = sqrt(2) S, and T ≈ 4.44288 / S.

Confirm.

- Measured T within 5e-6 relative of 2π/ω (Verlet phase error about $(\omega h)^2/24$).
- Max $|\Delta E/E|$ at most $(\omega h)^2 \approx$ 4e-5 with no secular growth.
- $|\Delta L/L|$ at most 1e-12 and R_P at most 1e-12.
- Δ^NA = 0.

(b) Two-body saturation

Setup. d = 10a, equal bodies, sweep τ_c. Expected F / F_thin = $S(\tau_c)^2$:

τ_c	S	S^2
0.1	0.96348	0.92829
1	0.70728	0.50024
10	0.14700	0.02161

Confirm.

- The ratio must be independent of d for d > a (exact identity). Test d/a = 1.5, 5, 50 to 1e-12 (closed form) and to 1e-10 with the Gauss-Legendre path forced on.
- Large-τ_c limit: F → ξ Π π a^4 / (4 d^2).
- Interior thin check: F = G_N m_j m_k d / a^3 to 1e-10.
- Continuity at d = a(1 ± 1e-9) to 1e-6.
- Reciprocity $|F_{12} + F_{21}|$ / |F| at most 1e-12.

(c) Non-additive screening

Setup. Three equal collinear spheres A, B, C with d_AB = d_BC = 10a, B in the middle. Define R(τ_c) = |F_A| / |F_AB + F_AC|.

Expected.

- R → 1 as τ_c → 0 (deviation O(τ_c)), and R decreases monotonically.
- As τ_c → infinity, R → 0.8 exactly: C′s disc lies fully inside B′s, so F_A = F_AB while the pair sum is 1.25 F_AB. At τ_c = 20, |R − 0.8| is at most 1e-6.
- F_B = 0 by symmetry, F_A = −F_C, and R_P at most 1e-9.

Confirm.

- Disjointness test: shift C laterally until the discs no longer overlap; then Δ^NA = 0 to 1e-13 (the correction list is empty).
- Independent check: a full-sky uniform-grid quadrature at N_side = 1024 must agree to 1e-3.
- Diagnostics: Δ^NA and R_P.

(d) Moving body in the flux (drag and aberration)

- **(d1) V1 drag.** Single body, W = 0, v_g / V_0 = 1e3, thin. Expected V(t) = V_0 exp(−γt) with γ = 16π G_N S / (3 κ_0 v_g), with relative correction $e^2/5$ ≈ 2e-7. The fitted γ must match to 1e-5.
- **(d2) Frame invariance.** The result must be identical to a body at rest with W = −V_0 (identical force).
- **(d3) V2 aberration.** Thin pair, |ε| about 1e-3. The measured force must match eq. 10 to O(ε^2) plus O(a^2/d^2) (target tolerance 10 ε^2). If the quadrature disagrees with eq. 10 beyond that, the quadrature governs and the discrepancy must be resolved before freezing.
- **Expected qualitative outcome.** Orbit decay at the rate γ in the drag-dominated regime.

(e) Additional required checks

- **Seeliger screening.** F(α_bg) / F(0) = exp(−α_bg d) [1 + O(α_bg a)] for α_bg a much less than 1.
- **Heating.** An isolated body satisfies P_heat = η_E ζ Π v_g A_eff exactly.
- **Equivalence principle.** Two bodies with $\delta_1 \neq \delta_2$ give η_EP = $|\delta_1 - \delta_2|$ to first order.

8. Explicit limitations of the numerical model

Inherited approximations

1. Static, non-relativistic, geometric-optics (ray) transport. Diffraction, wave effects and finite corpuscle wavelength are not modeled.
2. Center-sampled target. Attenuation is evaluated at x_k only. The error is O(a/d) in near-contact and overlap configurations, with exact shell-theorem behavior only at large separation. Overlapping spheres of different radii are not exactly reciprocal.
3. Scattered flux is removed from the beam, not re-injected (single-removal model). Error is O(ωτ) when scattering dominates.
4. Occluders are uniform spheres; real bodies′ internal structure and shape are ignored.
5. Midpoint quadrature in the correction: error O(f^(3/2)) at silhouette edges and O(ρ²) in vector weights at coarse cells. The adaptive tolerance caps out at ℓ_cap.
6. V1 drag is unshielded. V2 is first-order in ε for the reference formula and is restricted to α_bg = 0 and |ε| ⩽ 0.1.
7. The heating diagnostic is static-geometry only; relative-velocity corrections to energy absorption are ignored.
8. The background relaxation model (flux restored toward I_0) is phenomenological. It represents an assumed replenishment mechanism, not derived physics.
9. Finite-precision limits: τ_c above 1e8 (saturation precision), a/d below 1e-9 (disc angular resolution), and extreme v_g / V ratios.

Regimes where results are untrustworthy

- Cap-hit flagged configurations (very distant tiny overlapping occluders).
- Deep overlap with several spheres mutually inside each other (center-sampling error becomes O(1)).
- τ_c much greater than 1 combined with a heating interpretation (energy absorption diverges as v_g → infinity).
- v_g below 10 V_max (non-perturbative ε).
- Any use of the model to claim GR phenomena (precession consistent with GR, light bending, gravitational waves), cosmology, Lorentz covariance, or quantitative consistency with Puthoff′s or Sakharov′s mechanisms.
- Long-time energy conservation outside the thin, reciprocal, α_bg = 0 regime (non-conservation there is physical).

Implementation Contract

The implementation must compute, for every body and probe, the force of eq. 4 using the exact single-occluder pair forces of eqs. 5 and 6 (closed form A_{eff} / d^2 only for $\alpha_{bg} = 0$ exterior pairs, Gauss-Legendre otherwise) plus the full overlap-correction integral on the adaptive HEALPix scheme of section 4.3, evaluated in deficit form $D = 1 - $  with expm1 and the ordered ray-transfer recursion of eq. 17 when $\alpha_{bg} > 0$. It must apply the target self-shielding area A_{eff}, calibrate Π by eq. 19, integrate with eqs. 11 to 13, and expose every diagnostic of section 6.

It is forbidden to approximate away: the non-additive overlap correction; saturation ($S \neq 1$) of either occluder or target; finite sphere size (no point masses, no ad hoc Plummer softening); the deficit-versus-transmission formulation; or the calibration relation. It is likewise forbidden to substitute Barnes-Hut, fixed-count uniform ray sets, Monte-Carlo sampling, or thin-limit pairwise Newton for the full model in any conforming run. Thin-fast mode is the sole sanctioned deviation and must be labeled non-conforming.

Tests (a) to (e) must pass at the stated tolerances before the code is considered conforming. Any ambiguity found during implementation is to be resolved by the governing definitions (eqs. 3, 9, 16 and 17), not by implementer discretion.